

US Patent & Trademark Office

Patent Public Search | Text View

United States Patent	12413318
Kind Code	B2
Date of Patent	September 09, 2025
Inventor(s)	Beylin; Mikhail et al.

Over-the-air audio quality testing remote from testing site

Abstract

The technology includes a method, computer medium, and system for remote testing over-the-air (OTA) audio quality using a test platform positioned at core network, the test platform comprising first and second cellular handsets and bridging logic that interconnects, controls, and bridges the first and second cellular handsets. The method includes triggering the bridging logic. The bridging logic initiates a first call over the first cellular handset to a tester, and initiates a second call over the second cellular handset to a destination, wherein the second call is initiated over-the-air. The bridging logic and the first and second cellular handsets are positioned in a location that causes routing by a cellular network of the first call over a segment under test. The bridging logic bridging audio between the first and the second calls, including relaying audio. The tester determines that the second call was established based on the relayed audio.

Inventors: Beylin; Mikhail (Morgan Hill, CA), Royer; Ronald (Grapevine, TX), Phan; Dat (San Jose, CA), Bahjat; Ehab (Rockwall, TX), Seo; In (Fremont, CA)

Applicant: Spirent Communications, Inc. (San Jose, CA)

Family ID: 89430882

Assignee: Spirent Communications, Inc. (San Jose, CA)

Appl. No.: 18/127593

Filed: March 28, 2023

Prior Publication Data

Document Identifier	Publication Date
US 20240014909 A1	Jan. 11, 2024

Related U.S. Application Data

continuation-in-part parent-doc US 17858013 20220705 US 12200516 child-doc US 18127593

Publication Classification

Int. Cl.: H04B17/15 (20150101)

U.S. Cl.:

CPC H04B17/191 (20230501);

Field of Classification Search

CPC: H04B (17/191); H04W (24/06); H04M (3/2236); H04M (3/323); H04M (3/2281); G10L (25/60); G10L (25/69)

References Cited

U.S. PATENT DOCUMENTS

Patent No.	Issued Date	Patentee Name	U.S. Cl.	CPC
5551064	12/1995	Nobbe et al.	N/A	N/A
6201853	12/2000	Butler et al.	N/A	N/A
6337858	12/2001	Petty	370/410	H04M 7/003
6385646	12/2001	Brown	379/93.12	H04M 7/003
6577714	12/2002	Darcie	379/93.17	H04L 67/52
6728612	12/2003	Carver et al.	N/A	N/A
7181210	12/2006	Zabawskyj	455/433	H04W 88/14
7995516	12/2010	Zhou	455/445	H04W 92/02
8102986	12/2011	McClintock	370/352	H04M 1/72436
8150015	12/2011	Whitney	379/202.01	H04M 3/563
8868028	12/2013	Kaltsukis	N/A	N/A
9167079	12/2014	Capecci et al.	N/A	N/A
9398466	12/2015	Joung et al.	N/A	N/A
9774746	12/2016	Sennett	N/A	H04M 7/124
10097981	12/2017	Talaganov et al.	N/A	N/A
10212286	12/2018	Jayadevan	N/A	H04M 3/42008
10791221	12/2019	Vislocky	N/A	N/A
10805462	12/2019	Ginter et al.	N/A	N/A
10841413	12/2019	Pentakota et al.	N/A	N/A
2002/0131604	12/2001	Amine	381/58	H04R 29/004
2005/0043017	12/2004	Esposito	455/414.1	H04W 4/00
2007/0019769	12/2006	Green et al.	N/A	N/A
2007/0153088	12/2006	Hales et al.	N/A	N/A
2007/0177680	12/2006	Green et al.	N/A	N/A
2007/0291931	12/2006	DeMent	379/418	H04M 3/42017
2008/0037751	12/2007	Aldrey	379/202.01	H04M 3/56
2008/0095333	12/2007	Tudor	704/260	H04L 12/1818
2008/0096588	12/2007	Waytena	709/206	H04L 61/4535
2008/0159501	12/2007	Cai	N/A	N/A
2009/0003316	12/2008	Lee	370/352	H04L 12/5692
2009/0042539	12/2008	Jiang	455/408	H04M 15/00
2009/0144068	12/2008	Altberg	705/342	G06Q 30/02

2009/0147778	12/2008	Wanless	455/466	H04M 7/0054
2009/0207978	12/2008	Oldham et al.	N/A	N/A
2010/0048191	12/2009	Bender	455/414.1	H04M 15/00
2010/0091761	12/2009	Arnaud	370/352	H04L 12/66
2010/0135474	12/2009	Smith	N/A	N/A
2014/0344874	12/2013	Olsen	N/A	N/A
2016/0173693	12/2015	Spievak	379/265.09	G06Q 30/0269
2017/0104527	12/2016	Sinde	N/A	N/A
2017/0104713	12/2016	Agarwal	N/A	H04L 67/02
2017/0141845	12/2016	Totten et al.	N/A	N/A
2018/0149687	12/2017	Backus et al.	N/A	N/A
2019/0007806	12/2018	Chu	N/A	H04M 3/42059
2019/0116506	12/2018	Bendlin et al.	N/A	N/A
2019/0280438	12/2018	German et al.	N/A	N/A
2019/0280504	12/2018	Lee et al.	N/A	N/A
2019/0297503	12/2018	Traynor et al.	N/A	N/A
2019/0335298	12/2018	Mbonye et al.	N/A	N/A
2020/0035247	12/2019	Boyadjiev et al.	N/A	N/A
2021/0021979	12/2020	Chiang	N/A	N/A
2021/0233541	12/2020	Chen et al.	N/A	N/A
2023/0064462	12/2022	Santhar	N/A	H04M 3/562

OTHER PUBLICATIONS

3.5 mm Headset: Accessory Specification, Android Open Source Project, 5 pages, (“<https://source.android.com/devices/accessories/headset/plug-headset-spec#functions>” <accessed on Nov. 25, 2019>). cited by applicant

3.5 mm Headset Jack: Device Specification, Android Open Source Project, 4 pages, (“https://source.android.com/devices/accessories/headset/jack-headset-spec.html#software_mapping” <accessed on Nov. 25, 2019>). cited by applicant

Spirent Landslide™ E10 Data Sheet, Spirent Communications, Inc., Oct. 2018, 4 pages. cited by applicant

U.S. Appl. No. 16/733,126, filed Jan. 2, 2020, U.S. Pat. No. 10,841,413, Nov. 17, 2020, Granted. cited by applicant

U.S. Appl. No. 18/127,590, filed Mar. 28, 2023, 20240015245, Jan. 11, 2024, Published. cited by applicant

U.S. Appl. No. 18/127,595, filed Mar. 28, 2023, 20240015536, Jan. 11, 2024, Published. cited by applicant

“NENA i3 Standard for Next Generation 9-1-1, NENA-STA-010.3b-2021”, National Emergency No. Association, Inc., Oct. 7, 2020, 673 pages. cited by applicant

Green, “FCC Workshop E911 Phase II Location Accuracy Overview”, Sprint, 13 pages (downloaded from

(https://transition.fcc.gov/bureaus/pshs/911/Phase%202/Workshop_11_2013/Sprint_No2013_FINAL.pdf Sprint_No2013_FINAL)). cited by applicant

UserPlane Location Protocol v2.0.4, Section 5.1.15 Emergency Services Location Requests, Open Mobile Alliance, Dec. 13, 2018, pp. 107-112. cited by applicant

UserPlane Location Protocol v2.0.4, 6.1.5 Authentication Mechanisms applicable to an E-SLP, Open Mobile Alliance, Dec. 13, 2018, pp. 203-204. cited by applicant

FCC Fact Sheet, Implementing Kari's Law and Section 506 of Ray Baum's Act, Report and Order—PS Docket Nos. 18-261 and 17-239, and GN Docket No. 11-117, Jul. 11, 2019, 163 pgs. cited by applicant

Ray et al., Alternative solutions for new E911 rules in 2020, FreePBX.com (Sangoma Technologies), 5

pgs, (downloaded Mar. 16, 2022 from <https://community.freepbx.org/t/alternative-solutions-for-new-e911-rules-in-2020/64355>). cited by applicant

Public Law 115-127, 115th Congress, Amendment 47 USC Section 609—Kari's Law Act of 2017, Feb. 16, 2018, 2 pgs. cited by applicant

UserPlane Location Protocol v2.0.5, Section 5.1.12 Notification/Verification based on current location, Open Mobile Alliance, Oct. 28, 2019, pp. 88-89. cited by applicant

“The E-model: a computational model for use in transmission planning”, International Telecommunication Union (ITU), ITU-T Rec. G. 107, Jun. 2015, 30 pages. cited by applicant

Ray et al, Alternative solutions for new E911 rules in 2020, FreePBX.com (Sangoma Technologies), 5 pgs, (downloaded Mar. 16, 2022 from <https://community.freepbx.org/t/alternative-solutions-for-new-e911-rules-n-2020/64355>). cited by applicant

UserPlane Location Protocol v2.0.5, Open Mobile Alliance, Oct. 28, 2019, 403 pages. cited by applicant

Landslide E10 Datasheet, Spirent Communications, Inc., Oct. 2018, 4 pages. cited by applicant

“The E-model: a computational model for use in transmission planning”, Series G: Transmission Systems and Media, Digital Systems and Networks, Telecommunication Standardization Sector of ITU, G.107, Jun. 2015, 30 pages. cited by applicant

Primary Examiner: Rego; Dominic E

Attorney, Agent or Firm: Haynes Beffel & Wolfeld LLP

Background/Summary

RELATED APPLICATIONS (1) This application is a continuation-in-part of U.S. application Ser. No. 17/858,013, titled “E911 Backhaul Routing Tests Remote from Coverage Area,” filed on 5 Jul. 2022. (2) This application is related to the following contemporaneously filed applications: (3) U.S. application Ser. No. 18,127,590, titled “Transport Audio Quality Testing Remote From Testing Site,” filed 28 Mar. 2023; and (4) U.S. application Ser. No. 18,127,595, titled “Inter-Core Transport Audio Quality Testing Remote from Testing Site,” filed 28 Mar. 2023.

INCORPORATIONS BY REFERENCE

(1) The following are incorporated by reference for all purposes as if fully set forth here:

(2) U.S. application Ser. No. 16/733,126, Systems and Methods for Using an Audio Cross-connect Cable for Remote Mobile Device Testing, filed on 2 Jan. 2020, now U.S. Pat. No. 10,841,413 issued on 17 Nov. 2020.

(3) Spirent, “Data Sheet: Spirent Landslide™ E10—Small Form Factor, Mobile Core, Wi-Fi, Diameter, and IMS Test System,” date unknown [captured by Archive.org from [hxxps://www.spirent.com/products/core-network-test-5g-lte-ims-wifi-diameter-landslide](https://www.spirent.com/products/core-network-test-5g-lte-ims-wifi-diameter-landslide) on Nov. 11, 2019].

(4) Open Mobile Alliance (OMA), “UserPlane Location Protocol” (v2.0.5), published on 28 Oct. 2019, p 88-89.

(5) International Telecom Union (ITU) “The E-model: a computational model for use in transmission planning” ITU-T Rec. G.107, June 2015.

FIELD OF TECHNOLOGY

(6) The technology disclosed relates to audio testing in telecommunications, such as voice over long-term evolution (VoLTE), fourth-generation mobile networks (4G) and fifth-generation mobile networks (5G) for high-speed wireless communication for mobile phones. More specifically, the technology relates to remote testing of telephone transport. The test verifies the quality of transmitted voice.

ACRONYMS

(7) Acronyms used in this disclosure are identified the first time that they are used. These acronyms are terms of art, often used in standards documents. Except where the terms are used in a clearly and distinctly different sense than used in the art, we adopt the meanings found in the standards. For the reader's convenience, some acronyms are listed here:

(8) TABLE-US-00001 4G Fourth Generation wireless 5G Fifth Generation wireless A-GPS Assisted/Augmented GNSS CPE Consumer Premise Equipment DTMF Dual-Tone Multi Frequency H-SLP Home-SUPL Location Platform IMS IP Multimedia Subsystem KPI Key Performance Indicator LCC Local Connectivity Center LTE Long-Term Evolution MANO Management And Network Orchestration MLP Mobile Location Protocol MME Mobility Management Entity NFV Network Function Virtualization NFVi NFV Infrastructure NID Network Interface Device NMP Network Measurement Report NTE Network Telecommunications Equipment OMA Open Mobile Alliance ONT Optical Network Terminal OTA Over The Air QoP Quality of Position RF Radio Frequency RXLEV Received Signal Level SBC Session Border Controller SET SUPL Enabled Terminal SGW Serving Gateway SIP Session Initiation Protocol SLIR Standard Location Immediate Request SLIA Standard Location Immediate Answer SUPL Secure UserPlane Location TA Timing Advance UE User Equipment USB Universal Serial Bus VoLTE VOice over LTE VPN Virtual Private Network VTA Virtual Test Agent VTP Virtual Test Platform

BACKGROUND

(9) Telephone networks are becoming larger and increasingly complex. As former uninhabited areas are developed into towns and households, and as existing telephone networks in rural areas are increasingly upgraded to modern telephone infrastructure to support modern telephone capabilities, more segments of the transport infrastructure need to be tested.

(10) Before the advent of the disclosed technology, testing a particular transport segment for audio quality required a technician to be physically proximate to the transport segment under test. Therefore, test technicians (aka “telephone linesmen”) had to travel to the location served by an eNodeB and perform testing from there.

(11) Traffic jams, severe inclement weather conditions, and pandemics compound the cost of human travel to the transport segment and may make travel inadvisable. Certain locations may be inaccessible in practice (e.g., a military or sensitive R&D facility which restricts access may require a linesman to have a security clearance. Security clearance applications may take months to process).

(12) Also, it may be desirable to test other features of the telecommunication network, such as accuracy of location reporting (which may be useful for location-based services). These may necessitate different types of tests, which may require more time and effort by testers.

(13) An opportunity arises to provide remote testing of segments, eliminating the need for travel. Further, an opportunity arises to perform multiple tests of the transport segment in the same session.

Description

BRIEF DESCRIPTION OF DRAWINGS

(1) In the drawings, like reference characters generally refer to like parts throughout the different views. Also, the drawings are not necessarily to scale, with an emphasis instead generally being placed upon illustrating the principles of the technology disclosed. In the following description, various implementations of the technology disclosed are described with reference to the following drawings.

(2) FIG. 1 illustrates an example architecture for testing voice quality from a location without being

near that location.

(3) FIG. 2 illustrates potential Virtual Test Agent positions in a telephone system, depending on the level of Network Function Virtualization supported by the telephone network.

(4) FIG. 3 illustrates a use case of remotely testing audio quality over a transport segment associated with a core network.

(5) FIG. 4 illustrates a message flow diagram of the audio quality test

(6) FIG. 5 illustrates a use case of remotely testing audio quality of a transport segment between two core networks.

(7) FIG. 6 illustrates a message flow diagram of the audio quality test between two core networks.

(8) FIG. 7 illustrates a use case of remotely testing call delivery Over the Air.

(9) FIG. 8 illustrates a message flow diagram of the call delivery test.

(10) FIG. 9. Illustrates a timing diagram that explains the steps of providing positioning information via Secure UserPlane Location.

(11) FIG. 10 is a simplified block diagram that can be used to manage remote test agents, such as a VTA or testing platform.

DETAILED DESCRIPTION OF THE INVENTION

(12) The disclosed technology involves creating a media bridge between two calls—one call sent over the transport medium to be tested and another call sent to the tester. The media bridge is located in the core network, such as at a connectivity center.

(13) The disclosed technology permits audio testing to occur in minutes. By comparison, the traditional approach of telephone companies (sending linemen to test audio of a segment, waiting for the linesman to reach the location served by the eNodeB/cell tower, and prepare the testing equipment) can take hours. The technology accomplishes this time reduction by eliminating the need for travel to the physical transport segment itself. Rather, testing occurs remotely from an operations center, or even from the comfort of a tester's home. By eliminating travel times, the technology can facilitate completion of ten audio tests in the time required for the linesman testing by the traditional travel-test process.

(14) Briefly described, the technical system uses a bridge facilitator installed in a core network to initiate two separate direct calls to communication endpoints. The bridge facilitator then creates a media bridge between the two calls, and relays communications between the two calls. The bridge facilitator establishes a bi-directional audio path between the two communication endpoints.

(15) This technology provides control over the origination of calls that cannot be achieved using network conference calling. Traditional conference calling gives the carrier's network and call routing apparatus control over routing of three or more users. It is naive to expect that the user who joins two outgoing calls into a conference call would be relaying communications during the conference. Instead, the network decides how to route the communications. Moreover, conference calling introduces points of failure that would complicate evaluation of test results. So, the technology disclosed uses two calls and a custom audio bridge, instead of a conference call.

(16) The disclosure first provides a general environment, and then describes three use cases that align with the overall environment, using the example of audio testing. The disclosure then describes variations on tests and examples, and concludes with particular implementations of the technology.

(17) Architecture

(18) FIG. 1 illustrates an example architecture for testing audio from a location without being near that location.

(19) Telephone system **100** includes tester **141** at remote test facility **161**, transport **153**, bridging unit **155**, core network **165**, transport under test **157**, call endpoint **189** including options of test VTA **129**, operator **149**, user equipment (UE) **169**, call 1 **143**, call 2 **147**.

(20) Tester **141** may be a person that uses telecommunication software to perform voice quality tests, by sending audio to and receiving audio from call endpoint **189**. Alternatively, tester **141** may

be a bot that uses telecommunication software, and that generates audio signals as well as receiving audio from an entity at test call endpoint **189**. The received audio may be used to conduct audio quality tests, for example, by a person or as part of an E-model. E-model is described in ITU-T Rec. G.107, which is incorporated by reference herein.

(21) Remote test facility **161** can be a call center, or can be any other facility, such as a corporate office, or personal residence. The term “facility” is used loosely here (e.g., a bench at a public park in service proximity with a cellular tower may be a facility), and the phrase may even encompass a general area that can receive call 1 **143**. Remote test facility **161** is located remotely from transport under test **157**, and is served by transport **153**.

(22) Transport **153** may be any audio-bearing medium. When audio quality is being tested, transport **153** must also support audio of sufficient quality to avoid undue bias when evaluating the quality of call 2 **147**. To ensure audio quality tests provide usable results, “sufficient quality” of call 1 **143** audio may be evaluated using Key Performance Indicators (KPI), and/or techniques available in information theory.

(23) Bridging unit **155** is physically positioned on core network **165**. Bridging unit may be hardware test platform such as Spirent E10, software such as a Virtual Test Platform (VTP) or Virtual Test Agent (VTA), or a combination of hardware and software. Bridging unit **155** may represent a standalone unit, or may represent one of plural units positioned at multiple locations in core network **165**. Bridging unit **155** may record audio such as reference audio, transmitted audio and received audio, collect logs, post-process logs, as well as performing additional functions. Bridging unit **155** may support testing of Wi-Fi and IP Multimedia Subsystem (IMS) networks, including Over-The-Air (OTA) Long-Term Evolution (LTE) testing as well as mobile core testing. Bridging unit **155** is not limited to devices supporting specific wireless capabilities to connect to the cellular network, and instead represents the capability to connect and test independent of the particular radio technologies. Spirent Landslide is an example of a bridging unit. Two broad classes of Landslide are discussed below as examples.

(24) Core network **165** includes infrastructure controlled by an Evolved Packet Core (EPC). A core network **165** services an area with numerous cellular towers, such as a state. The EPC interacts with User Equipment (UE) through one of several eNodeBs, at cellular towers. The eNodeBs interacts with the EPC through a Mobility Management Entity (MME) and Serving Gateway (SGW). A large cellular network will have multiple (e.g., 50+) EPCs, and multiple MMEs per EPC.

(25) An MME is the control node for connection of cellular towers (eNodeBs) to a particular EPC in the cellular network. An eNodeB can be connected to multiple MMES, using a selection protocol. The MME is responsible for tracking and paging cell phones and other UE including retransmissions, and also for idle mode of User Equipment (UE). MME chooses the SGW for a UE during its initial attach.

(26) A SGW complements the MME to complete call setup and carry call traffic to and from the UE.

(27) Herein, core network **165** includes both elements of the telephone core network and elements that control the boundaries of the core network, such as edge routers, session border controllers, etc.). Thus, bridging unit may be positioned on boundary elements as well without falling outside of the contemplated disclosure.

(28) Irrespective of its position within a core network **165**, bridging unit **155** configures telephone system **100** to originate call 2 **147** from an area served by the EPC, over transport under test **157** test to call endpoint **189**.

(29) Transport under test **157** is a transport segment that is in the routing path of call 2 **147**. The reasons for testing the transport under test **157** may vary. As some examples, the reason may be to determine if transport under test **157** is cause of complaints for poor audio quality, or may be that transport under test **157** has been newly installed and a telephone company wishes to verify that transport under test **157** can complete a call, or may be part of a telephone company's periodic

maintenance.

(30) Call endpoint **189** may be a communication endpoint of call 2 **147** such as a designated test facility served by transport under test **157** or the location of a customer that is reporting audio quality issues.

(31) Call endpoint **189** may have one or more of: test VTA **129**, that may be configured to accept call 2 **147** and evaluate Key Performance Indicators (KPIs); an operator **149** who accepts call 2 **147** and reports subjective audio quality and/or other metadata about perceived audio quality; or UE **169** that is configured to automatically accept call 2 **147** and automatically evaluate audio quality KPIs. UE **169** may be a smartphone, a mobile phone, a desktop telephone, a wall-mounted telephone, computer equipment (e.g., a desktop computer or laptop) with telephone software installed, or other implementations.

(32) Testing for audio quality may involve the tester **141** transmitting voice and receiving transmissions of voice with call endpoint **189**. Testing for audio quality may also be automated using approaches such as E-model (which predicts the perceived “mouth-to-ear” audio quality based on telephony-band impairment factors, and determines a scalar quality rating value called Transmission Rating Factor).

(33) Testing may also involve reducing or simplifying other steps in call setup not provided here. For example, the test may ensure that the CODECS between tester **141** and call endpoint **189** are identical, removing the need for transcoding.

Example Bridging Unit: Landslide

(34) Landslide is a scalable platform to test and emulate 5G and O-RAN mobile networks built on traditional or cloud-native infrastructure. It generates real-world control and data plane traffic of millions of mobile subscribers as they move through the network. Virtualized test functions may be deployed directly on cloud infrastructure to better assess cloud-native network function performance. This eliminates the need for carriers to perform expensive, non-repeatable, and time-consuming drive testing in the live network.

(35) Landslide Virtual Test Agent

(36) FIG. 2 illustrates potential Virtual Test Agent positions in a telephone system, depending on the level of Network Function Virtualization supported by the telephone network.

(37) VisionWorks VTP Network Function Virtualization diagram **200** shows the various locations a VTA can be placed as the level of Network Function Virtualization (NFV) increases.

(38) VisionWorks is a software test platform that supports quality testing across telephony networks, at varying levels of NFV. Amongst its test agents, VisionWorks provides VTAs.

VisionWorks VTA is a catalog of microservices testing applications that are designed to perform tests with low computing resource costs and that can be deployed in multiple locations across the network. As the network's NFV stage increases towards total virtualization, more locations become available. Since VTAs are not physical testing components, they can be deployed while the tested system is online.

(39) For networks at the physical stage, a VTA may be installed at an edge router. For networks at the virtualization stage, a VTA may be installed on customer premises equipment, an edge router, core network, or data center on the internet. For networks at either of the orchestration stage and cloud-native stage, a VTA may be installed on consumer protection equipment, network terminating equipment (such as network interface devices and optical network terminals), access (such as fiber and ethernet), edge router, core network, and the internet.

(40) The position of the VTA does not affect its functionality. A VTA can control an EPCs even if not installed at that EPC. Thus, even large carrier with 50+EPCs can perform testing using VTAs installed at one or two EPCs.

(41) Landslide Test Platform

(42) When creating media bridges for testing, an alternative to VTAs is using a physical test platform.

(43) For example, the Spirent Landslide E10 is a small form factor test platform that supports applications for benchtop, lab, and distributed testing of mobile and Wi-Fi networks. The E10 can be itself be used as a standalone system that is connected with the tele-communication network, or can be a test server in connection with other test platforms (e.g., the Spirent C100).

(44) The E10 can support testing functions such as emulating users roaming and handing over throughout the mobile network, validating physical and virtual network functions for Mobile Core and IMS nodes, emulated adjacent network interfaces and nodes such that complete network topologies can be created on a desktop, Over the Air (OTA) testing over Mobile Radio Frequency (RF) network access devices using an OTA testing module (such as a U1 LTE module that can be installed on the E10) or real handsets, and other testing functionality. The E10 may have built-in tests or facilitate such tests. For example, the E10 can measure voice quality using Perceptual Objective Listening Quality Analysis (POLQA) Mean Opinion Score (MOS) in certain configurations, and can also facilitate E-model testing.

(45) When real handsets are attached, the E10 uses cross-connection technology to create an audio bridge. The particulars of cross-connection technology are described in U.S. application Ser. No. 16/733,126, Systems and Methods for Using an Audio Cross-connect Cable for Remote Mobile Device Testing, filed on 2 Jan. 2020, now U.S. Pat. No. 10,841,413 issued on 17 Nov. 2020, which is incorporated by reference.

(46) Three use-case examples of the technology applied to remotely audio testing of transport are provided below.

(47) Use Case #1: Remotely Testing Transport Segment

(48) FIG. 3 illustrates a use case of remotely testing audio quality over a transport segment associated with a core network.

(49) Telephone system **300** includes tester **321** named Tony, tester appliance **332**, Virtual Private Network (VPN) **323**, Session Initiation Protocol (SIP) server **314**, call manager **324**, core network **326**, Virtual Test Agent (VTA) **316**, transport segment under test **328**, operator **319**, User Equipment (UE) **329**, tester terminus **339**, call 1 **315**, and call 2 **317**.

(50) For example, consider tester Tony, who is at a laptop with a tester appliance located in Texas. Tony leverages a VTA that can connect to MME's across the country to approximate point-to-point customer connections. A customer who is experiencing an audio quality problem is in New York. The customer had audio quality problems while calling a friend in Pennsylvania. The carrier operates a pair of core networks cover New York and Pennsylvania, using a pair of EPCs. The EPCs each are connected to multiple eNodeBs.

(51) Tony would like to test the audio quality for calls between New York and Pennsylvania, even though he is in Texas and cannot travel to the East coast. Tony operates his test appliance two make two separate calls. The tester's test appliance uses a network connection to connect with a VTA in Texas, on the tester's own EPS (though the technology is not so limited—the VTA could be located anywhere in the core network) to connect a first call through an MME in Pennsylvania to himself, and a second call to the customer in NY who has called in their complaint. The connection using the MME in PA to the customer in NY approximates the call routing about which the customer called to complain.

(52) Telephone system **300** is a system with transport segment under test **328** that requires audio quality testing.

(53) In general, tester **321** and tester appliance **332** may be within a local area network (such as ethernet network serving a call center or corporate office) and connects to core network **326**, by Session Initiation protocol (SIP) server **314** managed by call manager **323**, through VPN **122**. Tester **321** may be a person. Tester appliance **332** may be hardware (e.g., a telephone) or software (e.g., a SIP client executing on a computer). In some implementations, tester appliance **332** may be configured to relay audio from tester **321** or alternatively may be configured to both relay audio from tester **321** and generate audio usable to test audio quality. In further implementations,

generated audio may be prerecorded audio and/or may be generated by a computer program. In our example, tester Tony and the test appliance he is using are in Texas.

(54) Tester **321** and tester appliance **332** are typically remote from transport segment under test **328**. Tester **321** and tester appliance **332** could be physically separated from transport segment **328** past touching distance, by at least 1 mile apart, 10 miles, 30 miles, 50 miles, or even physically located in entirely different administrative districts (e.g., different United States counties) that are served by a core network. In our example, a Tony in Texas is servicing a complaint from the East Coast, regarding a connection between New York and Pennsylvania.

(55) Generally, VTA **316** may be provisioned in core network **326** as an eNodeB, and originates calls from core network **326** by emulating a subscriber making a call from the eNodeB. VTA **316** is capable of emulating one or more UE on any sector of any cell tower through any core. The emulated UE may initiate calls, including call 1 **315** and call 2 **317**.

(56) VTA **316** controls call origination by sending signaling traffic to the MME of core network **326**. Once the call is established, the data is sent to an SGW. In this way, it is unnecessary for the signaling traffic initiating the test to be sent through an eNodeB. A network connection to VTA **316** is sufficient to deliver the signaling traffic to the MME and initiate the call.

(57) Tester terminus **339** is a communication endpoint for a test call. Tester terminus **339** may be a call center, a private domicile, or other building containing operator **319**, that may be communicatively reached by audio routed over transport segment under test **328**. Tester Terminus **339** may also facilitate the customer reporting the problem. Tester terminus **339** may instead be, or contain, User Equipment **329** that automatically or semi-automatically respond to a testing request.

(58) The test process may be initiated when VTA **316** receives a signal to test transport segment under test **328**. VTA **316** selects tester **321** to perform the test. The signal may be of any form and from any source (e.g., an out of band signal from tester **321** requesting test initiation, test allocation by a transport segment test dispatcher (not shown) to an available tester **321**, a computer program that interfaces with VTA **316** that randomly assigns tester **321**, etc.)

(59) Use Case #1 Call Flow

(60) FIG. 4 illustrates a message flow diagram for an audio quality test of a transport segment associated with a core network.

(61) Message flow diagram **400** shows how tester appliance **322**, VTA **316**, and tester terminus **339** communicate during the audio quality test to tester terminus **339**. In our hypothetical example, tester appliance **322** is with the service representative in Texas and the tester terminus is the customer being serviced in New York.

(62) At step A, VTA **316** makes a direct call 1 **315** to tester appliance **332**, and waits for call 1 **315** to be established.

(63) At step B, VTA **316** confirms that call 1 **315** has been established.

(64) At step C, VTA **316** sends an RTP echo to tester appliance **332**. This step avoids the network dropping call 1 **315** while setting up call 2 **317**.

(65) At step D, once call 1 **315** to tester appliance **332** has been established, VTA **316** makes direct call 2 **317** to transport terminus **339**. VTA **316** routes call 2 **317** over transport segment under test **328**.

(66) At step E, VTA **316** confirms that call 2 **317** has been established.

(67) At step F, once the second call to testing terminus **339** is established, VTA **316** bridges the first and second calls, thus configuring the system to relay audio between the two calls.

(68) At step G, once the audio bridge is established, audio quality of transport segment under test **328** is tested by relaying audio between tester appliance **332** and testing terminus **339** via the audio bridge. Audio testing may involve, for example, tester **321** exchanging verbal communication with operator **319**, and tester **321** may report subjective impressions of audio quality or cause measurement of audio quality KPIs.

(69) Use Case #2: Remote Testing Transport Between Core Networks

(70) FIG. 5 illustrates a use case of remotely testing audio quality of a transport segment between two core networks.

(71) Telephone system **500** includes tester **521**, tester appliance **522**, VPN **523**, SIP server **514**, call manager **524**, transport segment **528** being tested, call 1 **515**, call 2 **517**, VTA A **516**, VTA B **519**, core network A **526**, and core network B **529**.

(72) For example, consider that tester Tony and his laptop, who is still located in Texas, and the New York customer's friend is still in Pennsylvania. In this scenario, the customer and friend are served by two different core networks. Here, the test is to determine whether the transport between the core networks has negatively impacted the audio sent between the two core networks.

(73) Again, Tony would like to test the audio quality for calls between New York and Pennsylvania, even though he is in Texas and cannot travel to the East coast. Tony operates his tester appliance to make two calls, by signaling an MME in Pennsylvania. The first call is to himself, in TX, and the second is to the customer in NY who has called in their complaint. The connection through the MME in PA to the customer in NY approximates the call routing about which the customer called to complain.

(74) Telephone system **500** is the system with transport segment **528** being tested. Transport segment **528** undergoing test may be between core network A **526** and core network B **529**.

(75) In this example, tester **521** has the task of testing the audio quality of transport segment **528** that sits between core network A **526** and core network B **529**. Tester **521** and tester appliance **522** may be within a local area network (such as ethernet network serving a call center or corporate office) and connects to core network **526** by SIP server **514** and managed by call manager **524** through VPN **122**. Tester appliance **522** may be hardware (e.g., a telephone) or a combination of hardware and software (e.g., a SIP client executing on a computer).

(76) Tester **521** and tester appliance **522** may be remote from transport segment **528**. For example, tester **521** and tester appliance **522** could be physically separated from transport segment **528** that exceeds touching distance, at least 1 mile apart, 10 miles, 30 miles, 50 miles, or even physically located in entirely different administrative districts (e.g., different United States counties) that are served by core networks A **526** and B **529**.

(77) Generally, VTA A **516** may be provisioned in core network A **526** as an eNodeB, and originates calls from core network A **526** by emulating a subscriber making a call from the eNodeB. VTA A **516** may be capable of emulating UE in any Local Connectivity Center (LCC), such as New York, without the physical presence of tester **521**. An LCC is a physical location with people and equipment. The emulated UE may initiate calls, including call 1 **515** and call 2 **517**. In some implementations, the emulated UE is emulating more than one UE. In such implementations, call 1 **515** and call 2 **517** are made from separate emulated UE. In other implementations, the emulated UE making call 1 **515** and call 2 **517** is a single emulated phone. VTA **516** controls call origination by sending signaling traffic to the MME of core network **326**. Once the call is established, the data is sent to an SGW. In this way, it is unnecessary for the signaling traffic initiating the test to be sent through an eNodeB. A network connection to VTA **316** is sufficient to deliver the signaling traffic to the MME and initiate the call.

(78) VTA B **519** may be installed on core network B **529**, and serves as a communication endpoint for call 2 **517**. VTA B **519** may emulate UE in an LCC other than the first LCC, such as in Pennsylvania. VTA B **519** may be configured to record and report audio quality metrics, responsive to being contacted as part of an audio quality test. VTA B **519** may alternatively be installed at the customer reporting the problem.

(79) The test process may be initiated when VTA A **516** receives a signal to test transport segment **528**, which is between the example of Pennsylvania and New York. VTA A **516** selects tester **521** to perform the test. The signal may be of any form and from any source (e.g., an out of band signal from tester **521** requesting test initiation, test allocation by a transport segment test dispatcher (not shown) to an available tester **521**, a computer program that interfaces with VTA A **516** that

randomly assigns tester **521**, etc.) Tester **521**, operating test appliance **522**, evaluates the call.

(80) Use Case #2 Call Flow

(81) FIG. **6** illustrates a message flow diagram of the audio quality test between two core networks.

(82) Message flow diagram **600** shows how tester appliance **522**, VTA A **516**, and VTA B **519** communicate during the audio quality test between two core networks.

(83) At step A, VTA A **516** makes a first direct call **515** to tester appliance **522**, and waits for first call **515** to be established.

(84) At step B, tester appliance **522** confirms establishment of first call **515**.

(85) At step C, sends an RTP echo to tester appliance **332**. This step avoids the network dropping call 1 **315** while setting up call 2 **317**.

(86) At step D, once first call **515** to tester appliance **522** has been established, VTA A **516** makes direct call 2 **517** to VTA B **519**. VTA A **516** makes call 2 **517** routed over transport segment **528**.

(87) At step E, VTA A **516** confirms that call 2 **517** has been established to VTA B **519**.

(88) At step F, once call 2 **519** to VTA B **519** is established, VTA A **516** bridges both call 1 **515** and call 2 **517**, thus configuring the system to relay audio between the two calls.

(89) At step G, once the audio bridge the calls is established, audio quality is tested by relaying audio between tester appliance **522** and VTA B **529** via the audio bridge. In some implementations, VTA B **529** may contain a prerecorded message for tester **521**, at tester appliance **522**, to perform subjective or objective (such as by KPI) evaluation of audio quality. In some implementations, VTA B **529** may also record KPIs based on audio relayed from tester appliance **522**, and report the KPIs to tester appliance **522**.

(90) Use Case #3: Remote Testing of Over the Air

(91) FIG. **7** illustrates a use case of remotely testing call delivery Over the Air.

(92) Telephone system **700** includes test platform **715**, User Equipment (UE) “A” **724**, UE “B” **726** connected by audio cross-connect **725**, Customer Support “C” **731**, Destination “D” **739**, audio medium **723**, and OTA medium **727**. Customer support C **731** is also referred to as tester.

(93) The test itself determines if a call can be delivered OTA from UE B **726** to D **739**.

(94) Test platform **715** is configured with bridging logic that is suitable to test a telephone connection made Over the Air (OTA), and is located at a core network that serves destination **739**. Test platform **715** is connected to UE A **724** and UE B **726** by Universal Serial Bus (USB) connections. As an alternative or addition to a USB connection, test platform **715** may be configured with a Bluetooth™ component, such as a built-in component or a USB adapter, that connects with UE A **724** and UE B **726**. In this particular example, test platform **715** is a Spirent Landslide™ E10 Small Form Factor, Mobile Core, Wi-Fi, Diameter, and IMS Test System, and is configured to perform OTA testing. Particular features of the E10 can be found in the Spirent Landslide™ E10 datasheet, which is incorporated by reference in its entirety. Test platform **715** is not limited to Landslide. Other test platforms capable of OTA testing may also be used as test platform **715** or may be configured to be used as test platform **715**. Examples of those other test platforms include those by Keysight, Rohde & Schwarz, etc.

(95) UE A **724** and UE B **726** may be cellular handsets that are linked by audio cross-connect **725**, enabling transmission of audio between UE A **724** and UE B **726**. In some implementations, the UE may be laptops, handheld transceivers, or other portable electronic communications devices that are capable of interfacing with the test platform **715**.

(96) Customer support C **731** may be a person that tests whether calls can be delivered OTA to Destination D **739**. C **731** may be physically located remotely from test platform **715** and Destination D **739**. Remote location could be outside of physical touching distance, a distance of a mile, a distance of 10 miles, a distance 50 miles, or even having locations within different administrative districts. The location of C **731** is not limited, other than being remote from test platform **715**.

(97) Destination D **739** is a telephone being used to test whether audio sent OTA reaches the

location of D 739. In some implementations, D 739 may be operated by customers that are reporting low audio quality to a telephone system company. In other implementations, D 739 may be operated by one or more testing professionals whose responsibilities encompass testing local audio.

(98) In some implementations, audio medium 723 may be air or include air. In other implementations, audio medium 723 may be any other transport medium used to transmit telephone signals. In any variation of this use case, OTA medium 727 is the link under test. OTA medium 727 includes air.

(99) Test platform 715 initiates the audio test using bridging logic. The bridging logic of test platform 715 calls customer support C 731 from UE A 724 via audio medium 723.

(100) For the test to be successful, C 731 must hear and know that the call completed. Neither KPIs measurement nor DTMF is necessary.

(101) Use Case #3 Call Flow

(102) FIG. 8 illustrates a message flow diagram of the call delivery OTA test.

(103) Message flow diagram 800 shows how UE A 724, UE B 726, Customer Support C 731, and Destination D 739 communicate during the call delivery test.

(104) At step A, a first call between UE A 724 and C 731 is initiated, over audio medium 723

(105) At step B, the first call between UE A 724 and C 731 is established.

(106) At step C, two actions occur by UEB 726 at the same time. The first action is that a second call between 726 and D 739 is initiated over OTA medium 727. The second action is that audio cross-connect 725 is activated, this establishing a bi-directional audio bridge between UE A 724 and UEB 726. This permits A 724 to receive the audio from the second call setup between UEB 726 and D 739 being established.

(107) For purposes of this example, “at the same time” for step C may be within the span of 1 second. In general, part of the test is for UE A 724 to receive the audio from the call setup between UEB 726 and D 739. Typical call setup time for VoLTE, as of 2023, may be between 3-5 seconds. It is contemplated that as future technological advances reduce average call setup time, the threshold for “at the same time” may also be commensurately reduced such that UE A 724 can hear the call setup between UE B 726 and D 739.

(108) Once both calls over audio medium 723 and OTA medium 727 are established, audio may be relayed between C 731 and D 739 via audio cross-connect 725 between UE A 724 and UE B and 726.

(109) At step D, the second call is established. C 731 can hear that the second call was established via the bi-directional audio bridge.

(110) At step E, C 731 speaks and the spoken audio is relayed via medium 723 to A 724. B 726 receives the audio through audio cross-connect 725 with A 724. B 726 then relays the audio OTA to D 739.

(111) At step F, D 739 speaks a response. The response audio is sent OTA through OTA medium 727 to B 726, through audio cross-connect 725 to A 724, and then through audio medium 723 to C 731.

(112) At this point, the test is complete. Again, no measurement of KPIs is necessary for this particular test. It would be also an easy variation to also test audio quality, such as presented in the first two examples.

(113) The three uses cases above are examples of how the technology enables remotely testing transport. Each possible permutations and variations of the features amongst the examples that also enable remotely testing transport may not be expressly set forth in its own example, but would be understood by those of ordinary skill in the art, and are thus disclosed.

(114) Variations on Use Cases

(115) Common potential modifications to the use cases may involve measures to ensure the accuracy of the test. For example, when testing for audio quality, first call (whether call 1 315, call

1 515, or the first call over audio medium 723) may itself be evaluated for sufficient audio quality to avoid undue bias in determining the audio quality of the second call. Each use case may involve the bridging unit (whether the bridging unit is a VTA, a testing platform, or other such unit) sampling the first call to produce KPIs.

(116) Although the use cases, as described above, focus on audio quality testing, the technology is not limited to such. Any media capable of being exchanged through Real Time Protocol (RTP) may be bridged. For example, instead of an audio bridge, the technology could be extended to transmitted video. The technology also supports testing of other call properties, such as latency.

(117) Another advantage of the technology is that other tests can be conducted simultaneously with the quality test.

(118) Earlier-filed application E911 Backhaul Routing Tests Remote From Coverage Area, U.S. Ser. No. 17/858,013 (SPIR 1167-1), which revolved around E911 infrastructure and is the parent of this disclosure, involved testing location-based routing to particular public-safety answering points without having to send a mobile device operator to the actual physical location.

(119) Testing location-based routing may also be useful for non-emergency service contexts. For example, modern-day location-based services may rely on location information being transmitted during a call session from a mobile device for the customer's convenience when providing local customer service. As another example, such services may provide local purchasing-delivery opportunities that may be of interest while the customer is traveling through the local area. Thus, testing the correctness of location-based routing and accuracy of reported location may be of interest in non-E911 contexts.

(120) Verifying correct routing may also be used as a sanity check during call quality tests, as described above, to ensure the correct transport is being tested. Using use case #2 as an example, if the test of call quality that is ostensibly between the city of Miami and the city of New York reports crystal clear quality, but the connectivity center hosting VTA 519 serves Kansas, this may indicate further underlying problems.

(121) A variation of the disclosed technology, as deployed through non-E911 infrastructure, may facilitate tests of location-based routing.

(122) Secure UserPlane Location

(123) FIG. 9 illustrates a timing diagram that explains the steps of providing positioning information via Secure UserPlane Location. The positioning information can be sent to, and verified by, call endpoint 189.

(124) Timing diagram 900 illustrates how location information is obtained and provided to call endpoint 189. This is an edited description from Open Mobile Alliance (OMA), UserPlane Location Protocol v2.0.5 p88-89. The entirety of the UserPlane Location Protocol specification is incorporated by reference. Edits to the description include, without any intended change to meaning, padding capital letter message names with quotes to permit the reader to distinguish over the use of capital letter abbreviations and correction of minor typographical errors.

(125) At step A, the Secure UserPlane Location (SUPL) Agent issues a Mobile Location Protocol (MLP) "SLIR" message to the Home-SUPL Location Platform (H-SLP), with which SUPL Agent is associated. The H-SLP authenticates the SUPL Agent and checks if the SUPL Agent is authorized for the service it requests, based on the client-id received. Further, based on the received ms-id the H-SLP applies subscriber privacy against the client-id.

(126) If a previously computed position which meets the requested Quality of Position (QoP) is available at the H-SLP and, based on that position, no notification or verification is required, the H-SLP directly proceeds to step J. If, based on that position, notification and verification or notification only is required, the H-SLP proceeds to step B.

(127) At step B, The SLP verifies that the target terminal is currently within the service area of the SLP, i.e., the target terminal is not roaming. The SLP may also verify that the target terminal supports SUPL.

(128) At step C, The H-SLP initiates the location session with the SET using the SUPL "INIT" message. The SUPL "INIT" message contains at least session-id, proxy/non-proxy mode indicator and the intended positioning method. As in this case the result of the privacy check in Step A indicates that subscriber privacy check based on current location is required, the H-SLP includes the Notification Mode element in the SUPL "INIT" message to indicate notification based on current location and does not include the notification element in the SUPL "INIT" message.

(129) Before the SUPL "INIT" message is sent the H-SLP also computes and stores a hash of the message.

(130) If in step A the H-SLP decided to use a previously computed position, the SUPL "INIT" message indicates this in a 'no position' posmethod parameter value and the SUPL Enabled Terminal (SET) responds with a SUPL "END" message carrying the results of the verification process (access granted, or access denied). If no explicit verification is required (notification only) the SET responds with a SUPL "END" message. The H-SLP then directly proceeds to step J.

(131) At step D, the SET analyzes the received SUPL "INIT." If found to be non-authentic, the SET takes no further actions. Otherwise, the SET takes needed action preparing for establishment or resumption of a secure connection.

(132) At step E, the SET will evaluate the Notification rules and follow the appropriate actions. The SET checks the notification mode indicator and determines that in this case the notification is performed based on the location of the SET. The SET also checks the proxy/non-proxy mode indicator to determine if the H-SLP uses proxy or non-proxy mode. In this case, proxy mode is used, and the SET establishes a secure connection to the H-SLP using H-SLP address that has been provisioned by the Home Network to the SET. The SET then sends a SUPL "POS INIT" message to start a positioning session with the H-SLP. The SUPL "POS INIT" message contains at least session-id, SET capabilities, a hash of the received SUPL "INIT" message (ver) and Location ID (lid). The SET capabilities include the supported positioning methods (e.g., SET-Assisted Assisted-GNSS (A-GPS), SET-Based A-GPS) and associated positioning protocols (e.g., RRLP, RRC, TIA-801 or LPP/LPPe). The SET may provide a Network Measurement Report (NMR) specific for the radio technology being used (e.g., for GSM: Timing Advance(TA), Received Signal Level (RXLEV)). The SET may provide its position, if these are available and supported by both SET and H-SLP. The SET MAY set the Requested Assistance Data element in the SUPL "POS INIT".

(133) If a position received from or calculated based on information received in the SUPL "POS INIT" message is available that meets the required QoP, the H-SLP may directly proceed to step G and not engage in a SUPL POS session.

(134) At step F, checks that the hash of SUPL "INIT" matches the one it has computed for this particular session. Based on the SUPL "POS INIT" message including posmethod(s) supported by the SET the H-SLP then determines the posmethod. If required for the posmethod the H-SLP uses the supported positioning protocol (e.g., RRLP, RRC, TIA-801 or LPP/LPPe) from the SUPL "POS INIT" message.

(135) The SET and the H-SLP exchange several successive positioning procedure messages.

(136) The H-SLP calculates the position estimate based on the received positioning measurements (SET-Assisted) or the SET calculates the position estimate based on assistance obtained from the H-SLP (SET-Based).

(137) At step G, the H-SLP applies subscriber privacy against the SET position estimate determined in Step F. If, based on this position, notification and verification or notification only is required, the H-SLP sends a SUPL "NOTIFY" message to the SET. The SUPL "NOTIFY" message contains the notification element. If, based on this position, no notification and verification is required, the H-SLP directly proceeds to Step I.

(138) At step H, the SET sends a SUPL "NOTIFY RESPONSE" message to the H-SLP. If notification and verification was required in step G then this will contain the notification response from the user.

(139) At step I, once the position calculation is complete the H-SLP sends the SUPL “END” message to the SET informing it that no further positioning procedure will be started and that the location session is finished. The SET releases the secure connection to the H-SLP and release all resources related to this session.

(140) At step J, the H-SLP sends the position estimate back to the SUPL Agent in an MLP “SLIA” message and the H-SLP releases all resources related to the session.

(141) The following table provides the timing requirements shown in timing diagram **400**. UT refers to SET timers, and “ST” refers to SLP timers.

(142) TABLE-US-00002

Timer	Default value	Description	Expiration
UT2	11 seconds	From sending of SUPL POS For immediate applications the INIT to receipt of first SUPL SET sends SUPL END to the POS, SUPL REPORT or SUPL SLP and clears all session END message. UT2 is not resources. For triggered needed if the SUPL POS INIT applications, the SET skips the message contains the first SUPL POS session and SUPL POS element. continues the triggered session. UT3	10 seconds
UT3	10 seconds	From sending of the last SUPL For immediate applications, the POS message to receipt of SET sends SUPL END to the SUPL END, SUPL REPORT or SLP and clears all session SUPL NOTIFY. In cases where resources. For triggered there is no SUPL POS message applications, the SET continues sent from SET, timer UT3 is the triggered session. not used. UT5	10 seconds
UT5	10 seconds	Only applicable to “notification The SET sends SUPL END to based on location” scenarios. the SLP. From sending of SUPL The SET clears all session NOTIFY RESPONSE to receipt resources. of SUPL END ST2	10 seconds
ST2	10 seconds	for proxy, For proxy mode: from sending For non-roaming scenario: 50+ optionally response of SUPL INIT to receipt of Inform SUPL agent that the time in QoP SUPL POS INIT, SUPL session has ended. TRIGGERED START or SUPL For roaming scenario: Inform END. SUPL agent or, where For non-proxy mode: from applicable, R-SLP that the sending SUPL INIT to (a) session has receipt of notification (internal ended. communication between SPC For proxy: Clear session and SLC) that SUPL POS INIT resources at SLP has been received, (b) receipt of For non-proxy: Clear session RLP-SSRP(SUPL END) from resources at SLC and send V-SLP, (c) receipt of SUPL internal communication to SPC TRIGGERED START, (d) to clear receipt of SUPL REPORT or session resources at SPC where (e) receipt of SUPL END. applicable. ST5	10 seconds
ST5	10 seconds	From sending SUPL NOTIFY Send SUPL END to SET. to receipt of SUPL NOTIFY Clear session resources at SLP. RESPONSE.	

(143) The SET and the H-SLP may exchange several successive positioning procedure messages.

(144) The applications of location-based routing of phone numbers should be apparent to those skilled in the art. As a fictitious example of one such application, recall that in recent history, alcohol consumption laws were relaxed in many states to permit public outdoor drinking in a health-safe environment. Assume a nationwide chain of drinking establishments partners with local delivery services in states where light outdoor drinking is permitted. When a potential customer on a casual walk is thirsty for an adult beverage, the customer can call 1-800-SND-BEER (Again, this is a fictitious example. This is not a real phone number.) If the customer is in an area where outdoor drinking is legal, the call and the user's location will be routed to the nearest establishment with an available beer delivery-person to complete the order. If the customer is not in such an area, the customer may be directed to the closest area for delivery (or the closest bar). The ability to test location-based routing ensures that businesses opportunities can be captured without the risk of costing the participating businesses their liquor licenses.

(145) Other commercial uses of location-based testing (e.g., facilitating accuracy of location-based advertising) may be extrapolated by the example.

(146) Other types of testing, and their uses to test telephone network issues, can be extrapolated from the use cases and variations.

(147) Computer System

(148) FIG. **10** is a simplified block diagram of a computer system **1000** that can be used to manage remote test agents, such as a VTA or testing platform. Computer system **1010** includes at least one

central processing unit (CPU) **1072** that communicates with a number of peripheral devices via bus subsystem **1055**. These peripheral devices can include a storage subsystem **1026** including, for example, memory devices and a file storage subsystem **1036**, user interface input devices **1038**, user interface output devices **1076**, a network interface subsystem **1074**, and data I/O **1078**. The input and output devices allow user interaction with computer system **1010**. Network interface subsystem **1074** provides an interface to outside networks, including an interface to corresponding interface devices in a communication network with other computer systems.

(149) In one implementation, bridging unit **155** can be communicably linked to the storage subsystem **1026** and the user interface input devices **1038**. User interface input devices **1038** can include a keyboard; pointing devices such as a mouse, trackball, touchpad, or graphics tablet; a scanner; a touch screen incorporated into the display; audio input devices such as voice recognition systems and microphones; and other types of input devices. In general, use of the term “input device” is intended to include all possible types of devices and ways to input information into computer system **1010**.

(150) User interface output devices **1076** can include a display subsystem, a printer, a fax machine, or non-visual displays such as audio output devices. The display subsystem can include an LED display, a cathode ray tube (CRT), a flat-panel device such as a liquid crystal display (LCD), plasma, organic light-emitting diode (OLED), a projection device, or some other mechanism for creating a visible image. The display subsystem can also provide a non-visual display such as audio output devices. In general, use of the term “output device” is intended to include all possible types of devices and ways to output information from computer system **1000** to the user or to another machine or computer system.

(151) Storage subsystem **1026** stores programming and data constructs that provide the functionality of some or all of the modules and methods described herein.

(152) Memory subsystem **1022** used in the storage subsystem **1026** can include a number of memories including a main random access memory (RAM) **1034** for storage of instructions and data during program execution and a read only memory (ROM) **1032** in which fixed instructions are stored. A file storage subsystem **1036** can provide persistent storage for program and data files, and can include a hard disk drive, a floppy disk drive along with associated removable media, a CD-ROM drive, an optical drive, or removable media cartridges. The modules implementing the functionality of certain implementations can be stored by file storage subsystem **1036** in the storage subsystem **1026**, or in other machines accessible by the processor.

(153) Bus subsystem **1055** provides a mechanism for letting the various components and subsystems of computer system **1010** communicate with each other as intended. Although bus subsystem **1055** is shown schematically as a single bus, alternative implementations of the bus subsystem can use multiple busses.

(154) Computer system **1010** itself can be of varying types including a personal computer, a portable computer, a workstation, a computer terminal, a network computer, a television, a mainframe, a server farm, a widely-distributed set of loosely networked computers, or any other data processing system or user device. Due to the ever changing nature of computers and networks, the description of computer system **1010** depicted in FIG. **6** is intended only as a specific example for purposes of illustrating the preferred embodiments of the present invention.

(155) Many other configurations of computer system **1000** are possible having more or less components than the computer system depicted in FIG. **10**.

(156) The preceding description is presented to enable the making and use of the technology disclosed. Various modifications to the disclosed implementations will be apparent, and the general principles defined herein may be applied to other implementations and applications without departing from the spirit and scope of the technology disclosed. Thus, the technology disclosed is not intended to be limited to the implementations shown, but is to be accorded the widest scope consistent with the principles and features disclosed herein. The scope of the technology disclosed

is defined by the appended claims.

(157) Particular Implementations

(158) Some particular implementations and features are described in the following discussion.

(159) Remote Testing

(160) In one implementation, a disclosed method testing transport audio quality irrespective of tester location has a virtual test agent (VTA) selecting a Mobility Management Engine (MME) corresponding to an eNodeB positioned at a physical location served by a core network, for routing a second call through a Serving Gateway (SGW) assigned by the MME for audio communication with a tester terminus over a transport segment under test, whereby a tester can evaluate audio quality over the transport segment under test. The implementation also the VTA make a first call to a tester appliance. The implementation also has the VTA make the second call, over the transport segment under test, to the tester terminus by signaling the MME, wherein the tester terminus includes a person or User Equipment implementing audio evaluation. The implementation also bridges, by the VTA, the first and second calls by relaying audio during a test of subjective audio quality, whereby the bridging assures that the relayed audio in the second call continues to be relayed over the transport segment under test and not re-routed by core network components.

(161) This method and other implementations of the technology disclosed can include one or more of the following features and/or features described in connection with additional methods disclosed. In the interest of conciseness, the combinations of features disclosed in this application are not individually enumerated and are not repeated with each base set of features.

(162) For some implementations, the tester appliance is remote from the VTA and the transport segment. In further implementations, the user and the tester appliance are separated from the VTA and the transport segment by more than 50 miles.

(163) For some implementations, the user and the tester appliance are connected to the VTA by a VPN and through a SIP server.

(164) For some implementations, the tester terminus includes a person that interacts with the tester appliance. In further implementations, the person sends audio to the tester appliance.

(165) For some implementations, the tester terminus includes a test response application that interacts with the tester conducting the test. In further implementations, the test response application sends audio to the tester appliance.

(166) For some implementations, the tester is a person that conducts an audio quality test. In further implementations, the relayed audio was generated by the person, and evaluated at the tester terminus.

(167) For some implementations, the relayed audio was generated by a computer program and evaluated at the tester terminus.

(168) Remote Core to Core Transport Testing

(169) In one implementation, a disclosed method of testing audio quality of a transport segment between two core networks includes a first virtual test agent (VTA) selecting a Mobility Management Engine (MME) corresponding to an eNodeB positioned at a physical location served by a first core network for routing a second call, through a Serving Gateway (SGW) assigned by the MME for audio communication with a tester terminus, from the first VTA to a second VTA at a second core network over a transport-segment-under-test that connects the first and second core networks, whereby a tester can evaluate audio quality over the transport-segment-under-test. The implementation also has the first VTA making a first call to a tester appliance.

(170) The implementation also has making the second call, over the transport-segment-under-test, to the second VTA implementing audio evaluation by signaling the MME. The implementation also bridging, by the VTA, the first and second calls by relaying audio during a test of subjective audio quality, whereby the relay through the SGW assures that the relayed audio in the second call is relayed over the transport segment under test the first VTA making a first call to a tester appliance. The implementation also has the first VTA making the second call, over the transport-segment-

under-test, to the second VTA implementing audio evaluation by signaling the MME. The implementation also bridges, by the first VTA, the first and second calls by relaying audio during a test of subjective audio quality, whereby the relay through the SGW assures that the relayed audio in the second call is relayed over the transport segment under test.

(171) This method and other implementations of the technology disclosed can include one or more of the following features and/or features described in connection with additional methods disclosed. In the interest of conciseness, the combinations of features disclosed in this application are not individually enumerated and are not repeated with each base set of features.

(172) For some implementations, the method further includes facilitating an audio call evaluation of the second call. In further implementations, the audio call evaluation of the second call involves determining a transmission quality rating by E-model. In further implementations, the tester appliance is operated by a human tester to evaluate audio quality. In even further implementations, the tester is geographically located in an area not served by the first core network and not served by the second core network.

(173) For some implementations, first VTA makes the first call from a first emulated User Equipment (UE) and makes the second call from a second emulated UE.

(174) Remote OTA Testing

(175) In one implementation, remote testing over-the-air (OTA) audio quality using a test platform positioned at core network, the test platform comprising first and second cellular handsets and bridging logic that interconnects, controls, and bridges the first and second cellular handsets. The implementation includes triggering the bridging logic and causing the bridging logic to initiate a first call over the first cellular handset to a teste. The implementation also includes the bridging logic initiating a second call over the second cellular handset to a destination, wherein the second call is initiated over-the-air. The implementation also includes triggering the bridging logic and causing the bridging logic to initiate a first call over the first cellular handset to a tester. The implementation also includes the bridging logic initiating a second call over the second cellular handset to a destination, wherein the second call is initiated over-the-air, wherein the bridging logic and the first and second cellular handsets are positioned in a location that causes routing by a cellular network of the first call over a segment under test. The implementation also includes the bridging logic bridging audio between the first and the second calls, including relaying audio. The implementation also includes determining, by the tester, that the second call was established based on the relayed audio.

(176) This method and other implementations of the technology disclosed can include one or more of the following features and/or features described in connection with additional methods disclosed. In the interest of conciseness, the combinations of features disclosed in this application are not individually enumerated and are not repeated with each base set of features.

(177) For some implementations, triggering the bridging logic establishes a bi-directional audio path from customer support to the destination.

(178) For some implementations, the relaying involves relaying audio from customer support to the destination, followed by a response of relaying audio from the destination to customer support.

(179) For some implementations, the tester is remote from the test platform. In some further implementations, customer support is also remote from the destination.

(180) For some implementations, the audio relayed between the first cellular handset and the tester is transmitted OTA.

(181) For some implementations, the first or the second cellular handset connected with the rest of the test platform through a USB connection. For other implementations, the first or the second cellular handset connected with the rest of the test platform through Bluetooth. For further implementations, the test platform may be configured with a Bluetooth component, comprising either built-in Bluetooth component or a USB adapter, that connects with the cellular handsets.

(182) For some implementations, initiating the second call and bridging audio occurs at the same

time. For some further implementations, “at the same time” is within 1 second. For some further implementations, “at the same time” is a time that permits the first cellular handset to hear the call setup between the second cellular handset and the destination.

(183) Location-Based Routing

(184) For some implementations, testing may also include a test for location-based routing. For some further implementations, testing may location-based routing may test Secure UserPlane Location services. For some even further implementations, testing routing may occur contemporaneously with audio testing.

(185) For some implementations, testing by a tester may including measuring the quality of a first call with Key Performance Indicators; and if the KPIs do not exceed a threshold level of quality, selecting another tester and initiating the first call with the another tester.

(186) For some implementations, the relayed media is media capable of being relayed by Real-time Transport Protocol.

(187) Other implementations may include a non-transitory computer readable storage medium storing instructions executable by a processor to perform a method as described above. Yet another implementation may include a system including memory and one or more processors operable to execute instructions, stored in the memory, to perform a method as described above.

Claims

1. A method of remote testing over-the-air (OTA) audio quality using a test platform positioned at core network, the test platform comprising first and second cellular handsets and bridging logic that interconnects, controls, and bridges the first and second cellular handsets, the method including: triggering the bridging logic and causing the bridging logic to initiate a first call over the first cellular handset to a tester; the bridging logic initiating a second call over the second cellular handset to a destination, wherein the second call is initiated over-the-air; wherein the bridging logic and the first and second cellular handsets are positioned in a location that causes routing by a cellular network of the first call over a segment under test; the bridging logic bridging audio between the first and the second calls, including relaying audio; and determining, by the tester, that the second call was established based on the relayed audio.
2. The method of claim 1, wherein triggering the bridging logic establishes a bi-directional audio path from the tester to the destination.
3. The method of claim 1, wherein the relaying involves relaying audio from the tester to the destination, followed by a response of relaying audio from the destination to the tester.
4. The method of claim 1, wherein the tester is remote from both the test platform and the destination.
5. The method of claim 1, wherein the audio relayed between the first cellular handset and the tester is transmitted OTA.
6. The method of claim 1, wherein initiating the second call and bridging audio occurs at the same time.
7. The method of claim 6, wherein at the same time is a time that permits the first cellular handset to receive call setup audio between the second cellular handset and the destination.
8. A non-transitory computer readable storage medium impressed with computer program instructions to remotely test over-the-air (OTA) audio quality using a test platform positioned at core network, the test platform comprising first and second cellular handsets and bridging logic that interconnects, controls, and bridges the first and second cellular handsets, the instructions, when executed on a processor, implement a method comprising: triggering the bridging logic and causing the bridging logic to initiate a first call over the first cellular handset to a tester; the bridging logic initiating a second call over the second cellular handset to a destination, wherein the second call is initiated over-the-air; wherein the bridging logic and the first and second cellular handsets are

positioned in a location that causes routing by a cellular network of the second call over a segment under test; the bridging logic bridging audio between the first and the second calls, including relaying audio; and determining, by the tester, that the second call was established based on the relayed audio.

9. The non-transitory computer readable storage medium of claim 8, wherein triggering the bridging logic establishes a bi-directional audio path from the tester to the destination.

10. The non-transitory computer readable storage medium of claim 8, wherein the relaying involves relaying audio from the tester to the destination, followed by a response of relaying audio from the destination to the tester.

11. The non-transitory computer readable storage medium of claim 8, wherein the tester is remote from both the test platform and the destination.

12. The non-transitory computer readable storage medium of claim 8, wherein the audio relayed between the first cellular handset and the tester is transmitted OTA.

13. The non-transitory computer readable storage medium of claim 8, wherein initiating the second call and bridging audio occurs at the same time.

14. The non-transitory computer readable storage medium of claim 13, wherein at the same time is a time that permits the first cellular handset to receive call setup audio between the second cellular handset and the destination.

15. A system including one or more processors coupled to memory, the memory loaded with computer instructions to remotely test over-the-air (OTA) audio quality using a test platform positioned at core network, the test platform comprising first and second cellular handsets and bridging logic that interconnects, controls, and bridges the first and second cellular handsets, the instructions, when executed on the processors, implement actions comprising: triggering the bridging logic and causing the bridging logic to initiate a first call over the first cellular handset to a tester; the bridging logic initiating a second call over the second cellular handset to a destination, wherein the second call is initiated over-the-air; wherein the bridging logic and the first and second cellular handsets are positioned in a location that causes routing by a cellular network of the second call over a segment under test; the bridging logic bridging audio between the first and the second calls, including relaying audio; and determining, by the tester, that the second call was established based on the relayed audio.

16. The system of claim 15, wherein triggering the bridging logic establishes a bi-directional audio path from the tester to the destination.

17. The system of claim 15, wherein the relaying involves relaying audio from the tester to the destination, followed by a response of relaying audio from the destination to the tester.

18. The system of claim 15, wherein the tester is remote from both the test platform and the destination.

19. The system of claim 15, wherein the audio relayed between the first cellular handset and the tester is transmitted OTA.

20. The system of claim 15, wherein initiating the second call and bridging audio occurs at the same time.
